Nebius x NVIDIA Global AI Hackathon
Best Apps and Agents · Devpost

Naqsha

نقشہ what families in Pakistan call a house plan

Describe a home or upload a drawing of one. Get a floor plan checked against your city's building rules, a 3D model you can walk through, and photos from every angle.

Agents on NVIDIA Nemotron · Nebius Token Factory

Pivot Analytix · Abdullah Moazzam, Saad Wali Muhammad Joiya, Usama Bin Maalik



A photo made from Naqsha's 3D model

The problem

Most homes start as a contractor's sketch

Weeks of drafting

Every draft and every change goes back and forth with a draftsman, by hand.

No 3D until it's built

The family commits to a house they have only seen as flat drawings.

Few bylaw experts

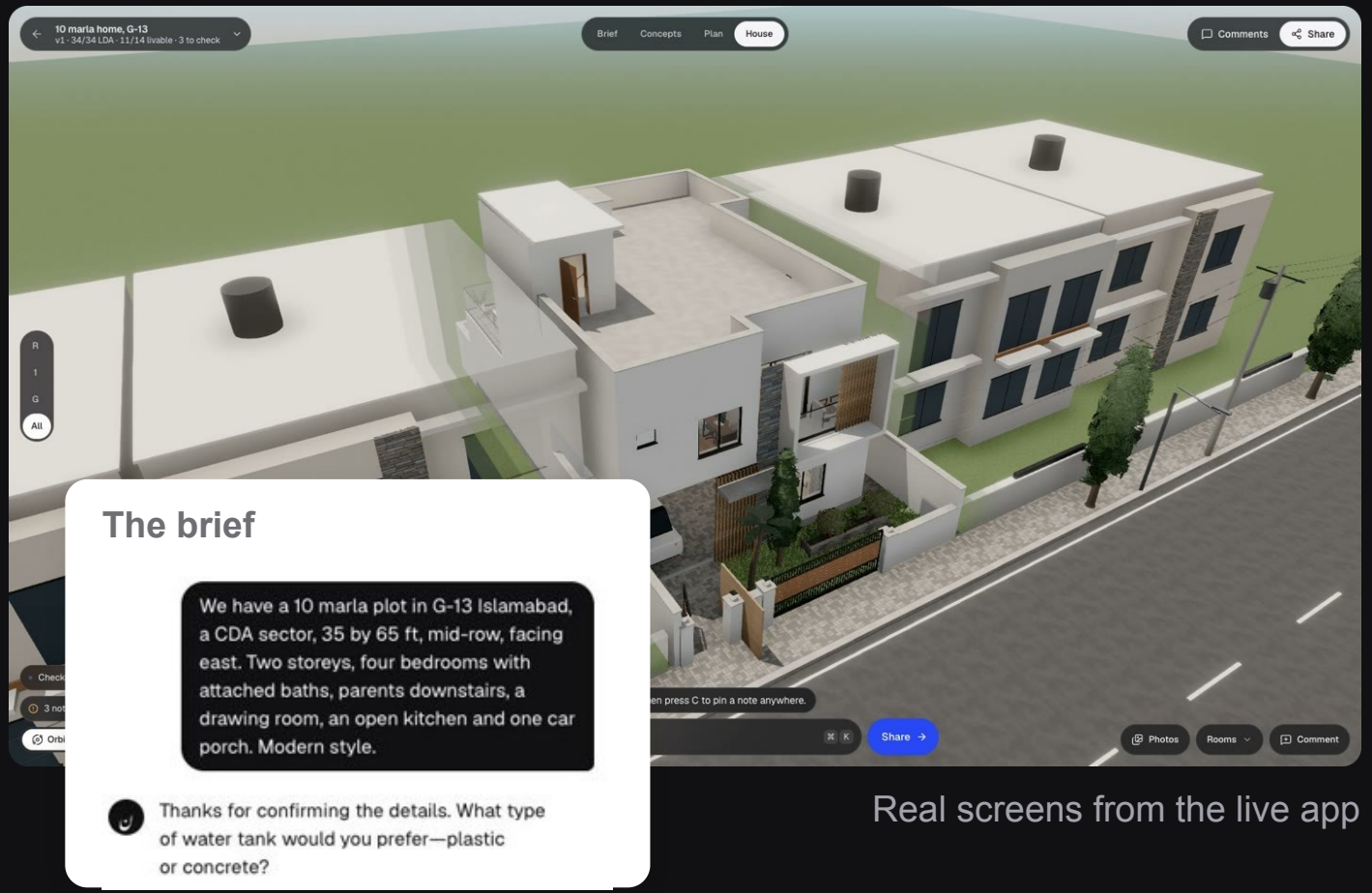
Compliance experts are scarce, so mistakes show up when the authority returns the plan.

Families in Pakistan usually build a house once. Nobody checks the plan against the rules until it is sent back.

What Naqsha does

From one message to a home you can walk through

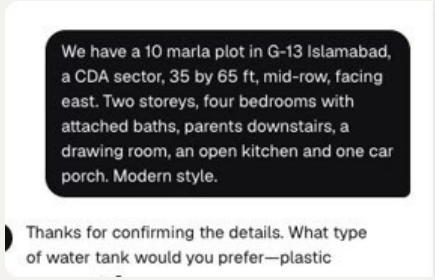
- A floor plan checked against your city's building rules
- A 3D model, written as code, that you can walk through
- Photos of the same house from every angle, and at night
- Comments from the family, applied by a revision agent
- Or start from a drawing you already have



Real screens from the live app

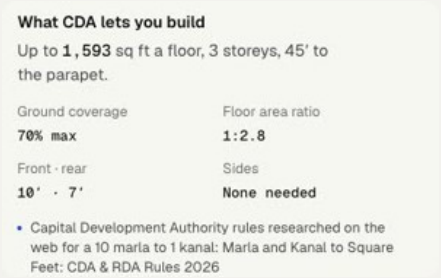
The journey

Nine steps from a brief to a walk-through



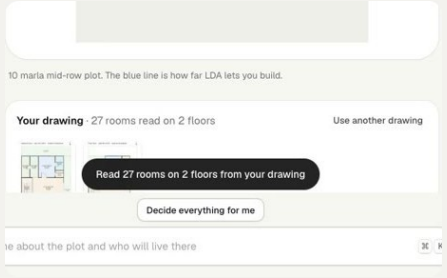
1 Brief

One message; the agent asks the rest



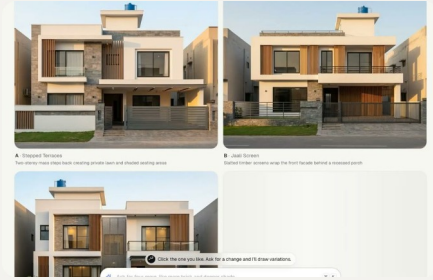
2 City rules

Looked up with Tavily, with sources



3 Your drawing

Optional; VL reads every room



4 Concepts

Four exteriors, by day and night



5 Plan + fixers

Four drafts, fixed down to zero failures



6 3D as code

Ultra writes the house in code



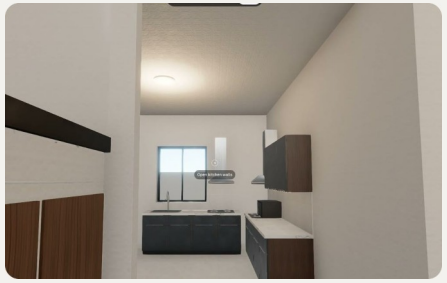
7 Concept match

Missing features get rebuilt



8 Photos

Four angles, made from the model



9 Walk

First person, room by room

Then share it, comment and revise

See slide 10

Start from your drawing

Upload the plan you have. Naqsha reads every room



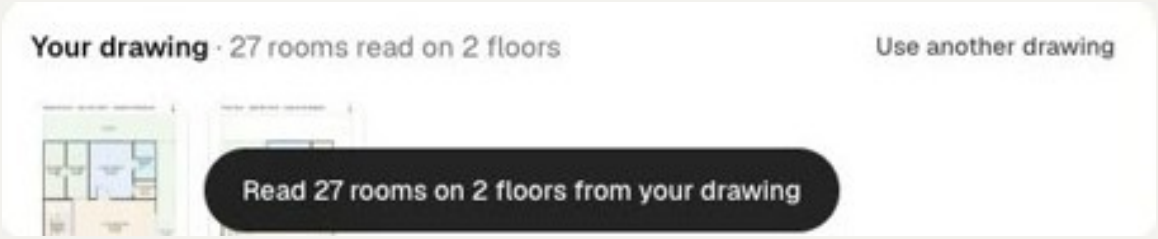
The two sheets uploaded in the demo

27 rooms

on 2 floors, read from two uploaded sheets

Nemotron 3.5 Super VL

It reads names, sizes, positions, doors and windows. The architect agent plans from that layout and changes only what the rules need, and the concepts are drawn from it.



The architect agent

How Nemotron 3 Ultra designs a home

Draft ×4

Nemotron 3 Ultra

Four plans in parallel, each adapted from a proven layout.

Check

Rules engine

Deterministic Python scores each draft against the bylaws.

Fix ×2

Nemotron 3 Ultra

Two fixer agents edit the best drafts with check_edits and apply_edits. The better plan wins.

Build

Nemotron 3 Ultra

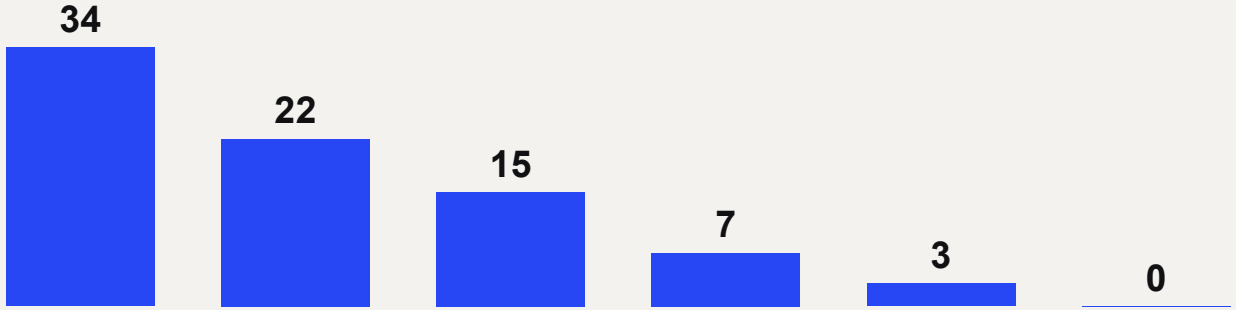
Writes the 3D model as code, section by section.

Match

Super VL + Ultra

VL compares the model with the concept. Ultra rebuilds what's missing.

Failing checks on the Islamabad plan, as the fixers worked





The plan view shows the countdown live. The finished plan passed all 34 bylaw checks.

Nebius Token Factory

Four Nemotron models, one endpoint

Nemotron 3 Ultra

The architect

- Four plan drafts from a proven layout
- Two fixer agents: check_edits, apply_edits
- The 3D model, written as code
- Rule packs from web pages
- Facade refits
- Revision agent (Google ADK, 9 tools)

Nemotron 3 Super

The conversation

- Setup agent (Google ADK, 6 tools)
- Records the brief and asks only what's missing

Nemotron 3.5 Super VL

The eyes

- Reads uploaded drawings
- Builds the concept feature checklist
- Checks the 3D model against the concept
- Reviews its own plan drawings

Nemotron 3.5 Lightning

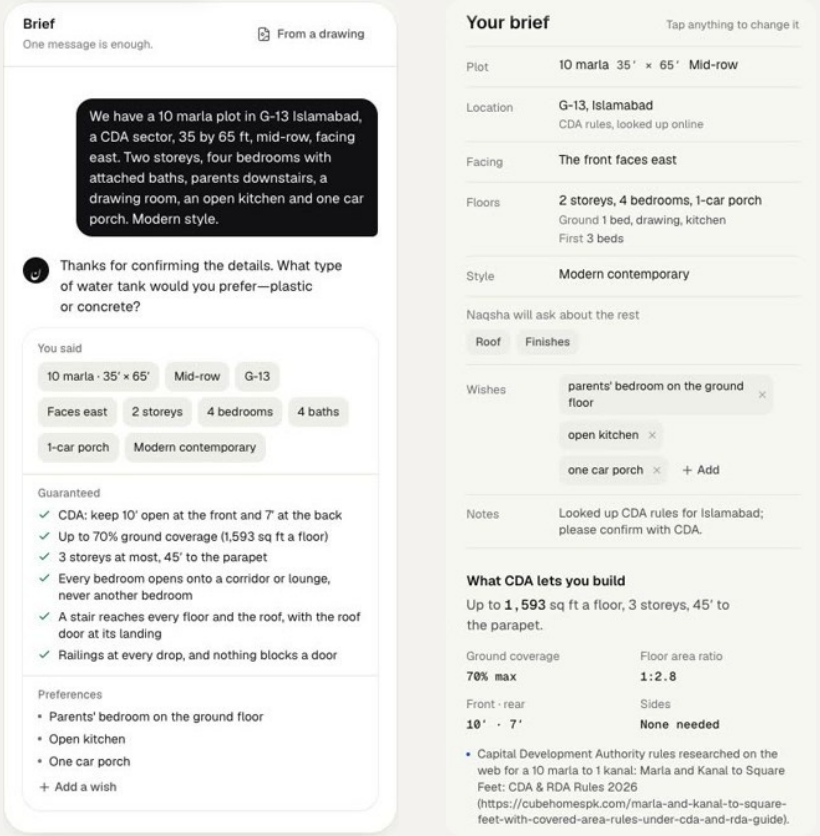
The fallback

- Takes over when a model is busy
- Ultra → Super → Lightning

Outside Token Factory: Tavily for web research, and Gemini image models for the concepts and photos.

Rules for any city

Building rules for any city, with sources



The Islamabad demo, in the live app

1. A brief names a plot outside Lahore, here G-13 in Islamabad.
2. Tavily Search finds the authority's residential rules. Tavily Extract pulls the pages, official sites first.
3. Nemotron 3 Ultra writes a rule pack: setbacks, coverage, FAR, height and storeys by plot size, each band citing its sources.
4. The rules engine checks the plan against that pack.

Lahore's LDA rules ship as a hand-verified pack.

The house

Built to match the concept, then photographed



93% of the concept is in the model, up from 86% before the facade was rebuilt. Nemotron VL checks feature by feature, and Ultra refits only the sections that miss.







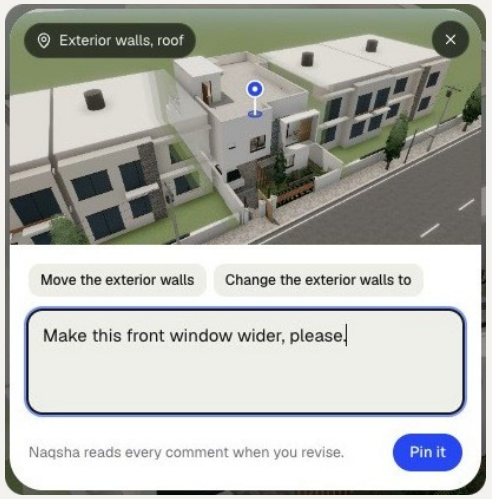


Photos from four angles. An image model turns each 3D capture into a photo without moving a wall. Every home also gets a night view and a walk-through.

Collaboration and revisions

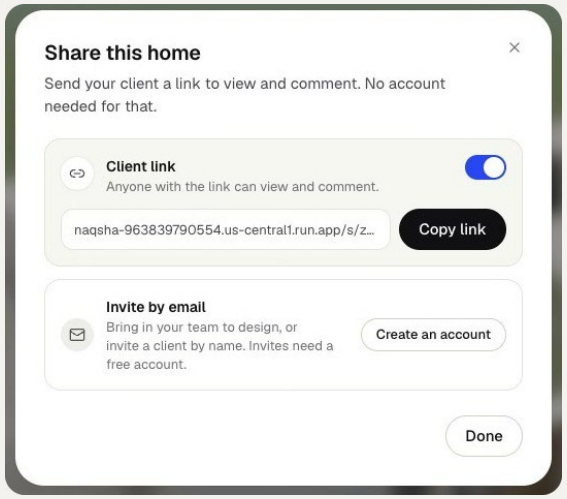
The family comments. The agent revises

1 Pin a note



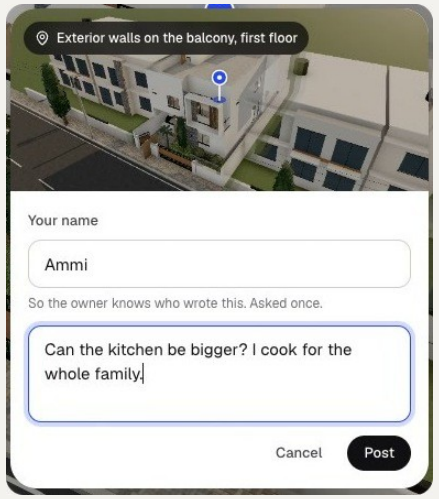
The owner pins one on the front window.

2 Share a link



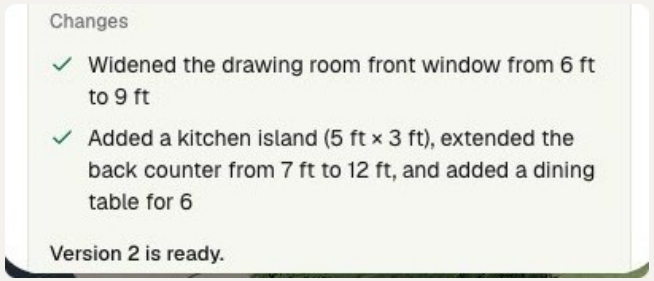
The family comments without an account.

3 Ammi asks



The client wants a bigger kitchen.

4 Revise



The revision agent applied both notes: a wider front window, and a kitchen island, a longer counter and a table for six.

A request that breaks a bylaw is refused, with the rule named.

Or type a change in the Ask bar, in plain words.

Demo runs · 10 marla homes

What the demo produced

34/34 bylaw checks pass on the Islamabad plan

93% of the concept's features are in the 3D model, up from 86%

27 rooms read on 2 floors from two uploaded sheets

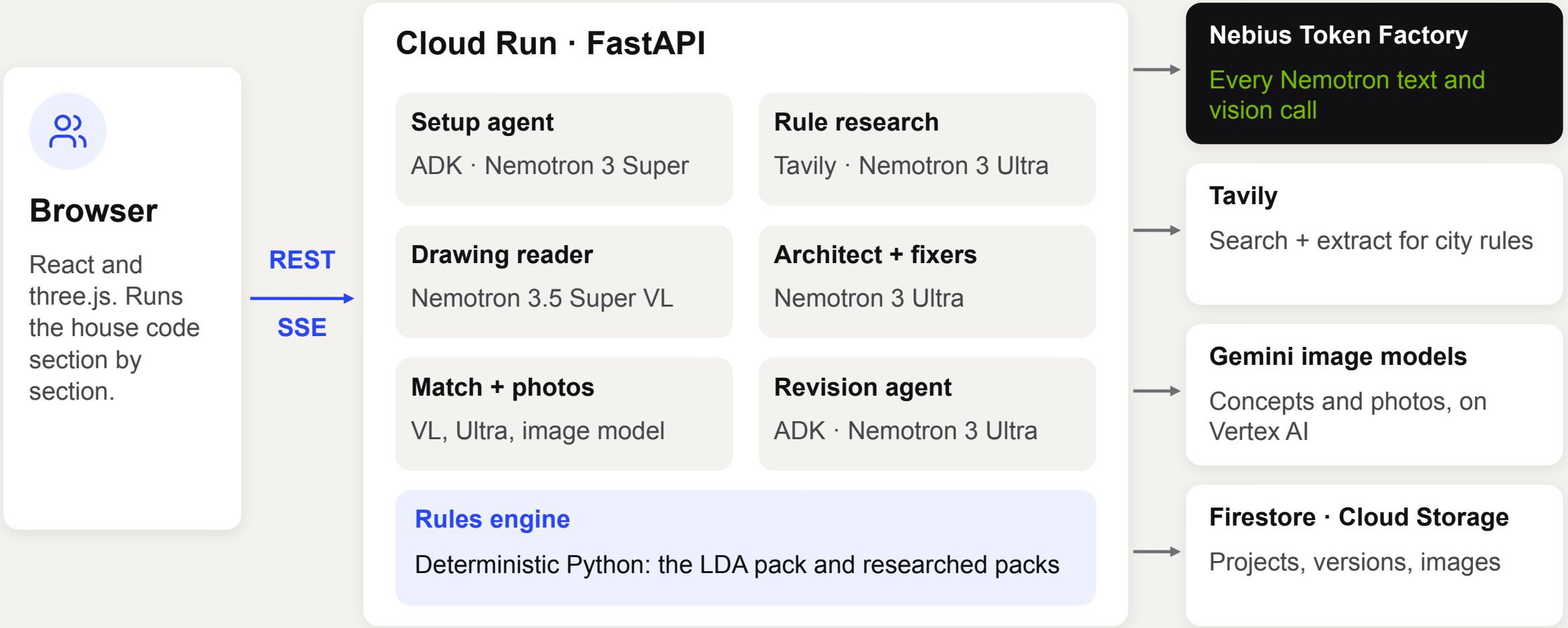


Fixers' countdown: 34 → 22 → 15 → 7 → 3 → 0 failing checks

Then: photos from 4 angles, a night view and a walk-through

Architecture

How the pieces fit



Who it's for

Who it helps, and how it scales

Families

See and walk through the house before spending on bricks. Comment from a shared link.

Architects

Bring a checked design to the first meeting, then revise it with the client.

Builders

Sell and build from designs that buyers can walk through.

Rules as data, so the same app scales city by city

Lahore
LDA pack, hand-verified, built in

→

Islamabad
CDA rules researched with Tavily in the demo

→

Pakistan
The other development authorities

→

Worldwide
Any city whose rules are published

Naqsha makes concept designs and a regulation pre-check. A registered architect still signs the plans, and researched rule packs should be confirmed with the authority.

What's next

What we're building next

Reviewed rule packs

Local architects review the packs for Karachi, Islamabad, Rawalpindi and Faisalabad.

Structural and MEP drafts

Structure, plumbing and electrical drafts from the same plan.

Bills of quantities

Quantities for every home, priced with local rates.

Nebius Serverless Jobs

The long design runs move off the web server and onto Serverless Jobs.



The goal: a family gets a checked plan, a costed house and a walk-through before the first meeting with a builder.

How to test

Try it now. No account needed



naqsha-963839790554
.us-central1.run.app

Free and open until judging ends.

1

Explore a sample home

Click **Explore a sample home**. It opens the Islamabad house. Try **Walk**, open **Photos**, and click the **Matches the concept** chip.

2

Decide everything for me

Click **Start your home** and type one message, e.g. “10 marla in G-13 Islamabad, 4 bedrooms, parents downstairs, modern”. Then click **Decide everything for me**. The house follows in about 10 minutes.

3

Start from your drawing

On the start page, click **Start from your drawing** and upload a photo of a floor plan. Nemotron VL reads it room by room.

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Pivot Analytix

Abdullah Moazzam lead
Saad Wali Muhammad Joiya
Usama Bin Maalik

Live app naqsha-963839790554.us-central1.run.app
no account needed

Code github.com/abdm693/naqsha
AGPL-3.0

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